Jan Škvrna

Computer Vision Researcher | 3D Perception, Generative 3D Models & Label-Free Learning

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PhD researcher in 3D computer vision at CTU Prague with four top-venue papers (**ECCV**, **ICCV**, **WACV**, **IJCV**) on label-free monocular 3D detection and generative 3D reconstruction, and two-time winner of the CVPR S23DR 3D reconstruction challenge; all methods released as open-source code. Three years of production embedded C/C++ before the PhD.

Selected Publications

WACV 2027 WireDiT: A Vertex-Slot Diffusion Transformer for 3D Wireframe Reconstruction. **J. Škvrna**, M. Purkrábek, L. Neumann — accepted for publication.

IJCV 2026 MonoSOWA++: Universal Autolabeling Pipeline for Monocular 3D Object Detection. **J. Škvrna**, L. Neumann. [Paper] [Code]

ICCV 2025 MonoSOWA: Scalable Monocular 3D Object Detector Without Human Annotations. **J. Škvrna**, L. Neumann. [Paper] [Code]

ECCV 2024 TCC-Det: Temporarily Consistent Cues for Weakly-Supervised 3D Detection. **J. Škvrna**, L. Neumann. [Paper] [Code]

Research Experience

PhD Researcher — 3D Computer Vision | Visual Recognition Group, CTU Prague | Feb 2024 – Present

Generative 3D wireframe reconstruction — S23DR Challenge, CVPR Urban Scene Modeling Workshop

- **WireDiT** (**WACV** 2027) — flow-matching DiT that denoises 64 vertex slots into a metric building wireframe from sparse street-level SfM, depth and semantics, conditioned on point-anchored Perceiver scene tokens, with hull-cropped second-stage refinement and multi-sample consensus. 68M params, 4×H200 DDP training, inference in seconds. State of the art on HoHo (HSS 0.654, vertex F1 0.791) and Building3D.
- **1st place, S23DR 2026** (**CVPR** 2026, $12k prize pool) — WireDiT's challenge version, developed end to end in a team of three. [Report] [Code]
- **1st place, S23DR 2025** (**CVPR** 2025, $25k prize pool) — native-3D two-stage PointNet pipeline for vertex refinement and edge prediction, developed end to end in a team of two. [Paper] [Code]

Training 3D detectors without target-domain human annotations

- **TCC-Det** (**ECCV** 2024) — auto-labelling pipeline that trains a LiDAR 3D detector with no target-domain human annotations from an off-the-shelf 2D instance segmentation model, temporal consistency across frames, and an outlier-robust Template Fitting Loss. New state of the art among annotation-free methods on KITTI and Waymo.
- **MonoSOWA** (**ICCV** 2025) — extended the approach to monocular video without target-domain human annotations, separating object from camera motion and normalising focal length across datasets. KITTI-360 pseudo-label pre-training plus 25% of KITTI labels outperformed the same detector trained from scratch on 100%.
- **MonoSOWA++** (**IJCV** 2026) — unified both into one pipeline at 1.9 s/frame (≈500× faster than the previous state of the art), with confidence-scored pseudo-labels and pedestrian/cyclist support.

Education

PhD in Informatics | CTU Prague, Visual Recognition Group, advisor Dr. Lukáš Neumann | Feb 2024 – Dec 2027 (exp.)

MSc in Cybernetics and Robotics, with Honours | CTU Prague, Dean's Award for Master's Thesis | 2021 – 2024

Industry & Teaching Experience

Junior Embedded Software Developer | Poll s.r.o. | Jun 2020 – Jun 2023

Real-time C/C++ firmware for ARM microcontrollers in HVAC and high-voltage DC/DC converter systems.

Lab Instructor — **Deep Learning Essentials** | CTU Prague | 2025 – Present

Academic Service & Training

Reviewer: ICCV 2025, BMVC 2026, AAAI 2027

Talks & Schools: CMP Colloquium, Prague (2026) | Computer Vision Winter Workshop, Graz (2025) | EEML (2026) | ICVSS (2025)

Technical Skills

Modelling: Diffusion & flow matching, DiT / Perceiver-style transformers, set prediction, 3D detection & structured reconstruction, multi-view geometry, depth & pose estimation, weak/self-supervised learning

Systems: Python, PyTorch (DDP, bf16, Slurm multi-GPU training), C/C++, CUDA, Linux, Git, OpenCV / Open3D / PyTorch3D, large-scale data processing pipelines